

INFOE 3: LAB 2_Data Swapping
Joy Melisa S. Segotier
Jorlyn Shane M. Claveria
Maria Theresa N. Reyes
BS INFO 3C

```
6 # Initialize the Faker generator
7 fake = Faker('en_US')
8
9 def generate_raw_data(num_entries=25): # Default is now 25
10     raw_data = []
11
12     for _ in range(num_entries):
13         entry = {
14             "name": fake.name(),
15             "age": random.randint(18, 85),
16             "zip_code": fake.postcode(),
17             "salary": random.randint(35000, 160000),
18             "medical_risk": random.choice(["Low", "Medium", "High"])
19         }
20         raw_data.append(entry)
21
22     return raw_data
23
24 # Generate 25 entries (or change to any number >= 25)
25 raw_data = generate_raw_data(25) # Change 25 to desired number
26
27 # Display the result
28 print(f'Generated {len(raw_data)} entries:\n')
29 pprint(raw_data[:3]) # Show first 3 records only
30
31 # Convert to DataFrame and save to CSV
32 df = pd.DataFrame(raw_data)
33 df.to_csv('employee_data.csv', index=False, encoding='utf-8-sig')
```

Generated 25 entries:

```
[{"age": 80,
  "medical_risk": 'Low',
  "name": 'Brian Jones',
  "salary": 44162,
  "zip_code": '50131'},
 {"age": 66,
  "medical_risk": 'Low',
  "name": 'Elizabeth Perez',
  "salary": 64680,
  "zip_code": '09495'},
 {"age": 34,
  "medical_risk": 'Low',
  "name": 'Nicole Gonzalez',
  "salary": 127257,
  "zip_code": '46683'}]
```

Dataset saved to 'employee_data.csv'
Total records: 25

This screenshot shows the generation of the raw_data list containing 25 unique entries. Each entry includes necessary attributes such as name, age, zip code, salary, and medical risk .

```
6
7 # Load the data
8 df = pd.read_csv('employee_data.csv')
9 raw_data = df.to_dict('records')
10
11 # Calculate Pre-Swap Metrics
12 original_salaries = [record['salary'] for record in raw_data]
13 pre_swap_total = sum(original_salaries)
14 pre_swap_avg = pre_swap_total / len(original_salaries)
15
16 print(f'Total Payroll: ${pre_swap_total:,.2f}')
17 print(f'Average Salary: ${pre_swap_avg:,.2f}')
18 print("-" * 60)
19
20 # Display sample of original data (first 3 records)
21 print("\nSAMPLE OF ORIGINAL DATA (Before Swap):")
22 for i in range(min(3, len(raw_data))):
23     print(f" {raw_data[i]['name']} - ${raw_data[i]['salary']:,.2f}")
24
25
26 # Extract salaries into a separate list
27 salary_list = [record['salary'] for record in raw_data]
28
29
30 def safe_shuffle(salary_list, original_salaries):
31     shuffled = salary_list.copy()
32     random.shuffle(shuffled)
33
34     shuffle_attempts = 1
35     while True:
36         # Check if any person got their original salary
37         any_same = False
38         for i in range(len(shuffled)):
39             if shuffled[i] == original_salaries[i]:
40                 any_same = True
41
42         if not any_same:
43             break
44
45     return shuffled
```

PS C:\Users\HYUN GADGETS\Desktop\Segotier, Claveria, Reyes_BS Info3C_Lab2_Info3> & "C:/Users/HYUN GADGETS/AppData/Local/Programs/Python/Python314/python.exe" "c:/Users/HYUN GADGETS/Desktop/Segotier, Claveria, Reyes_BS I nfo3C_Lab2_Info3/swapping_lab2.py"

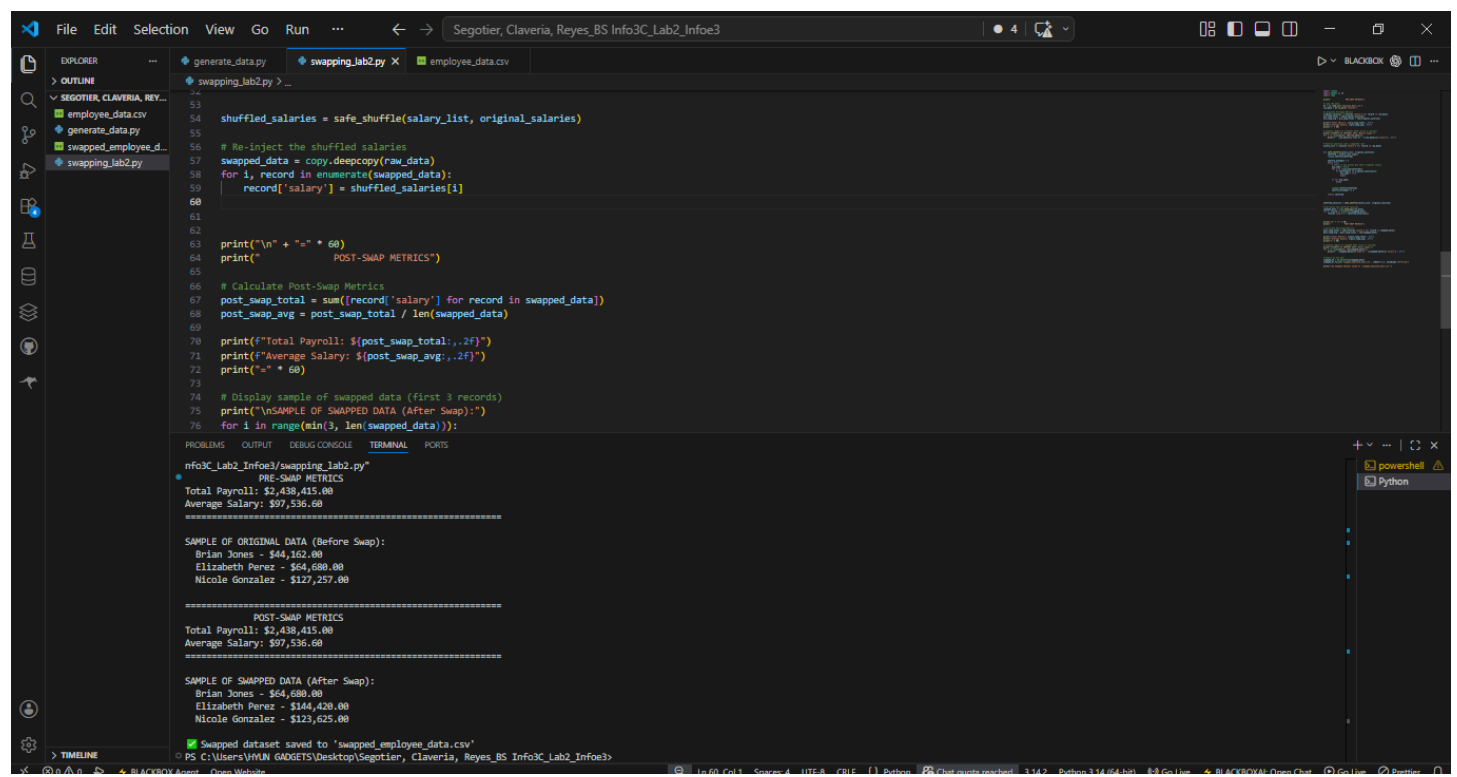
PRE-SWAP METRICS

```
Total Payroll: $2,438,415.00
Average Salary: $97,536.00
=====
```

SAMPLE OF ORIGINAL DATA (Before Swap):

```
Brian Jones - $44,162.00
Elizabeth Perez - $64,680.00
Nicole Gonzalez - $127,257.00
```

This image displays the "Pre-Swap" calculations for the dataset. It establishes the baseline Total Payroll and Average Salary, which are needed to verify statistical accuracy in the process.



The screenshot shows a Visual Studio Code editor with a Python script named `swapping_lab2.py` open. The script performs the following steps:

- Shuffles salaries from `employee_data.csv` into `shuffled_salaries`.
- Re-injects the shuffled salaries into the dataset, creating `swapped_data`.
- Calculates post-swap metrics: `post_swap_total` and `post_swap_avg`.
- Prints the total payroll and average salary after the swap.
- Displays a sample of the swapped data (first 3 records).

The terminal output confirms the success of the data swap by showing that the Total Payroll and Average Salary remain unchanged. It also displays a sample of the original data (before swap) and the swapped data (after swap).

```
python3 nfo3C_Lab2_Info3/swapping_lab2.py
PRE-SWAP METRICS
Total Payroll: $2,438,415.00
Average Salary: $97,536.60
=====
SAMPLE OF ORIGINAL DATA (Before Swap):
Brian Jones - $44,162.00
Elizabeth Perez - $64,680.00
Nicole Gonzalez - $127,257.00
=====
POST-SWAP METRICS
Total Payroll: $2,438,415.00
Average Salary: $97,536.60
=====
SAMPLE OF SWAPPED DATA (After Swap):
Brian Jones - $64,680.00
Elizabeth Perez - $144,438.00
Nicole Gonzalez - $123,625.00
=====
Swapped dataset saved to 'swapped_employee_data.csv'
```

This screenshot confirms the success of the data swap by showing that the Total Payroll and Average Salary remain unchanged. It demonstrates that the salaries have been redistributed while maintaining the dataset's overall statistical integrity.